

3rd Top Down Approach Lexical Analysis

Grammar 1

$\langle \text{assign} \rangle \rightarrow \langle \text{id} \rangle = \langle \text{expr} \rangle$
 $\langle \text{id} \rangle \rightarrow A \mid B \mid C$
 $\langle \text{expr} \rangle \rightarrow \langle \text{id} \rangle + \langle \text{expr} \rangle$
 $\quad \mid \langle \text{id} \rangle * \langle \text{expr} \rangle$
 $\quad \mid (\langle \text{expr} \rangle)$
 $\quad \mid \langle \text{id} \rangle$

1. Draw the parse trees for the following statements using grammar 1:

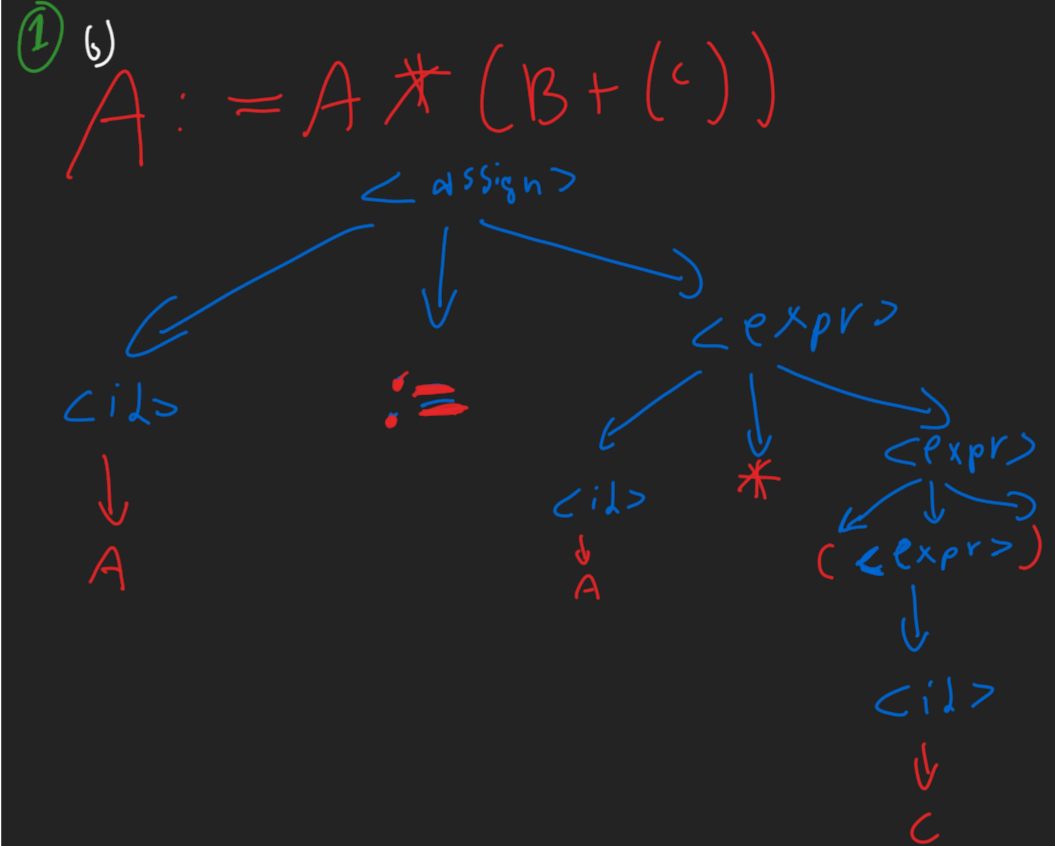
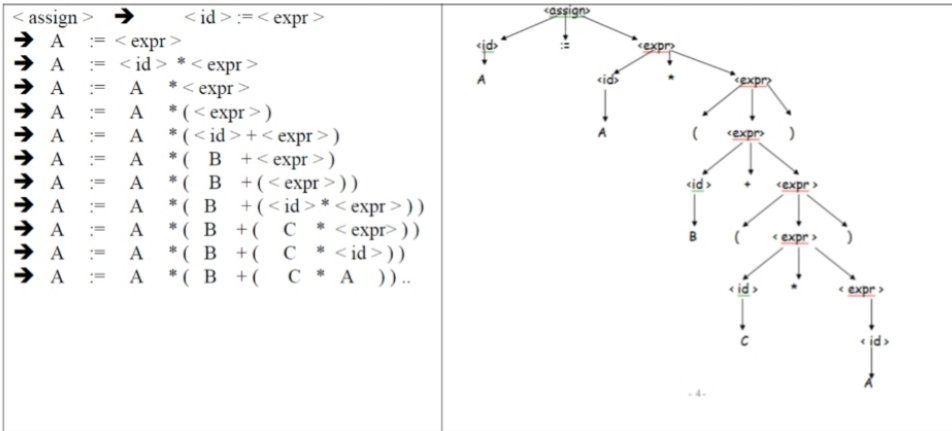
a. $A := A * (B + (C * A))$

b. $B := C * (A * C + B)$

c. $A := A * (B + (C))$

Solution

a.



$\langle \text{assign} \rangle \rightarrow \langle \text{id} \rangle := \langle \text{expr} \rangle$
 $\rightarrow A := \langle \text{expr} \rangle$
 $\rightarrow A := \langle \text{id} \rangle * \langle \text{expr} \rangle$
 $\rightarrow A := A * \langle \text{expr} \rangle$
 $\rightarrow A := A * (\langle \text{expr} \rangle)$
 $\rightarrow A := A * (\langle \text{id} \rangle + \langle \text{expr} \rangle)$
 $\rightarrow A := A * (B + \langle \text{expr} \rangle)$
 $\rightarrow A := A * (B + (\langle \text{id} \rangle))$
 $\rightarrow A := A * (B + (C))$

Youssef Eslam
231000070 Assignment 9

Grammar 2

$\langle \text{assign} \rangle \rightarrow \langle \text{id} \rangle = \langle \text{expr} \rangle$
 $\langle \text{id} \rangle \rightarrow A \mid B \mid C$
 $\langle \text{expr} \rangle \rightarrow \langle \text{expr} \rangle + \langle \text{term} \rangle$
 $\quad \mid \langle \text{term} \rangle$
 $\langle \text{term} \rangle \rightarrow \langle \text{term} \rangle * \langle \text{factor} \rangle$
 $\quad \mid \langle \text{factor} \rangle$
 $\langle \text{factor} \rangle \rightarrow (\langle \text{expr} \rangle)$
 $\quad \mid \langle \text{id} \rangle$

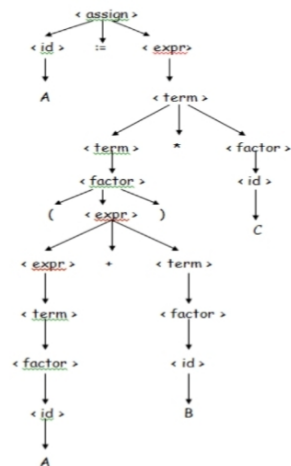
2. Draw the parse trees for the following statements using grammar 2:

- $A := (A + B) * C$
- $A := B + C + A$
- $A := A * (B + C)$

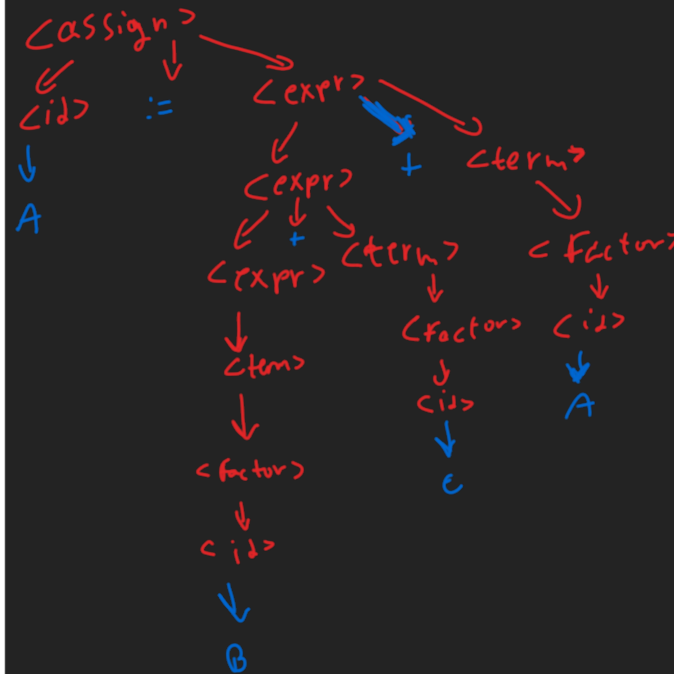
Solution

a.

$\langle \text{assign} \rangle \rightarrow \langle \text{id} \rangle := \langle \text{expr} \rangle$
 $\rightarrow A := \langle \text{expr} \rangle$
 $\rightarrow A := \langle \text{term} \rangle$
 $\rightarrow A := \langle \text{term} \rangle * \langle \text{factor} \rangle$
 $\rightarrow A := \langle \text{factor} \rangle * \langle \text{factor} \rangle$
 $\rightarrow A := (\langle \text{expr} \rangle) * \langle \text{factor} \rangle$
 $\rightarrow A := (\langle \text{expr} \rangle + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
 $\rightarrow A := (\langle \text{term} \rangle + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
 $\rightarrow A := (\langle \text{factor} \rangle + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
 $\rightarrow A := (\langle \text{id} \rangle + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
 $\rightarrow A := (A + \langle \text{term} \rangle) * \langle \text{factor} \rangle$
 $\rightarrow A := (A + \langle \text{factor} \rangle) * \langle \text{factor} \rangle$
 $\rightarrow A := (A + \langle \text{id} \rangle) * \langle \text{factor} \rangle$
 $\rightarrow A := (A + B) * \langle \text{factor} \rangle$
 $\rightarrow A := (A + B) * \langle \text{id} \rangle$
 $\rightarrow A := (A + B) * C$



② b) $A := B + C + A$



$\langle \text{assign} \rangle \rightarrow \langle \text{id} \rangle := \langle \text{expr} \rangle$

$\rightarrow A := \langle \text{expr} \rangle$

$\rightarrow A := \langle \text{expr} \rangle + \langle \text{term} \rangle$

$\rightarrow A := \langle \text{expr} \rangle + \langle \text{term} \rangle + \langle \text{term} \rangle$

$\rightarrow A := \langle \text{term} \rangle + \langle \text{term} \rangle + \langle \text{term} \rangle$

$\rightarrow A := \langle \text{factor} \rangle + \langle \text{term} \rangle + \langle \text{term} \rangle$

$\rightarrow A := \langle \text{id} \rangle + \langle \text{term} \rangle + \langle \text{term} \rangle$

$\rightarrow A := B + \langle \text{term} \rangle + \langle \text{term} \rangle$

$\rightarrow A := B + \langle \text{factor} \rangle + \langle \text{term} \rangle$

$\rightarrow A := B + \langle \text{id} \rangle + \langle \text{term} \rangle$

$\rightarrow A := B + C + \langle \text{term} \rangle$

$\rightarrow A := B + C + \langle \text{factor} \rangle$

$\rightarrow A := B + C + \langle \text{id} \rangle$

$\rightarrow A := B + C + A$